

Final Solutions

ECON 301

2022



Question 1

(a) The pdf is $f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}} I_{[0, \infty)}(x)$

Then:

$$E(X) = \int_0^{\infty} \frac{x}{\theta} e^{-\frac{x}{\theta}} dx = -x e^{-\frac{x}{\theta}} \Big|_0^{\infty} + \int_0^{\infty} e^{-\frac{x}{\theta}} dx$$

$$= - \left[\lim_{x \rightarrow \infty} \frac{x}{e^{\frac{x}{\theta}}} - 0 \cdot e^{-\frac{0}{\theta}} \right] - \theta e^{-\frac{x}{\theta}} \Big|_0^{\infty}$$

* Using Integration by parts:

$$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$$

Let $u = x$ $v = -e^{-\frac{x}{\theta}}$
 $du = dx$ $dv = \frac{1}{\theta} e^{-\frac{x}{\theta}} dx$

$$= -\theta \left(\lim_{x \rightarrow \infty} e^{-\frac{x}{\theta}} - e^{-\frac{0}{\theta}} \right) = -\theta(0 - 1) = \theta$$

$$E(X^2) = \int_0^{\infty} \frac{x^2}{\theta} e^{-\frac{x}{\theta}} dx = -x^2 e^{-\frac{x}{\theta}} \Big|_0^{\infty} + 2 \int_0^{\infty} x e^{-\frac{x}{\theta}} dx$$

Using integration by parts:

$u = x^2$ $v = -e^{-\frac{x}{\theta}}$
 $du = 2x dx$ $dv = \frac{1}{\theta} e^{-\frac{x}{\theta}} dx$

$$= 2\theta \int_0^{\infty} \frac{x}{\theta} e^{-\frac{x}{\theta}} dx = 2\theta^2$$

= θ from above

$$\therefore \text{Var}(X) = E(X^2) - (E(X))^2 = 2\theta^2 - \theta^2 = \theta^2$$

(b) Since the sample is i.i.d., the WLLN applies so that $\bar{X}_n \xrightarrow{p} E(X) = \theta$. Therefore $\bar{X}_n$ is consistent.

(c) Again, since the sample is i.i.d., the CLT applies; since $E(\bar{X}_n) = \theta$ and $\text{Var}(\bar{X}_n) = \frac{\theta^2}{n}$ we have that:

$$\frac{\bar{X}_n - \theta}{\theta/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

This implies that $\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{d} N(0, \theta^2)$

(d) Note that:

$$\begin{aligned} S_n^2 &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \frac{1}{n} \sum_{i=1}^n (X_i^2 - 2X_i \bar{X}_n + \bar{X}_n^2) \\ &= \frac{1}{n} \left[\sum_{i=1}^n X_i^2 - 2n \bar{X}_n^2 + n \bar{X}_n^2 \right] \\ &= \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2 \end{aligned}$$

Now: $E(\bar{X}_n^2) = \text{Var}(\bar{X}_n) + (E(\bar{X}_n))^2 = \frac{\theta^2}{n} + \theta^2 = \theta^2 \left(\frac{n+1}{n} \right)$

Also
$$E(X_i^2) = \text{Var}(X_i) + (E(X_i))^2 = \theta^2 + \theta^2 = 2\theta^2$$

Thus
$$E\left(\frac{1}{n} \sum_i X_i^2\right) = \frac{1}{n} \sum_i E(X_i^2) = \frac{2n\theta^2}{n} = 2\theta^2$$

Hence
$$E(S_n^2) = 2\theta^2 - \theta^2\left(\frac{n+1}{n}\right) = \frac{2n\theta^2 - \theta^2(n+1)}{n} = \left(\frac{n-1}{n}\right)\theta^2$$

Therefore, neither $\bar{X}_n^2$ nor S_n^2 are unbiased.

(e) By theorem 4.3, if we define $g(x) = x^2$

then $g(\bar{X}_n) \xrightarrow{p} g(\theta) = \theta^2$. Thus

$$\bar{X}_n^2 \xrightarrow{p} \theta^2 = \text{Var}(X)$$

Once again an i.i.d. sample implies we can apply the WLLN to the sequence:

$$\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{p} E(X^2) = 2\theta^2$$

Defining $h(z, w) = z - w^2$, we can invoke Thm 4.3

to get that

$$S_n^2 = h\left(\frac{1}{n} \sum_i X_i^2, \bar{X}_n\right) = \frac{1}{n} \sum_i X_i^2 - \bar{X}_n^2 \xrightarrow{p} 2\theta^2 - \theta^2 = \theta^2 = \text{Var}(X)$$

$\therefore$ both $\bar{X}_n^2$ and S_n^2 are consistent (for the variance)

(f) Since both $\bar{X}_n^2$ and S_n^2 are consistent estimators for $\text{Var}(X) = \theta^2$, it is OK to use any of them.

Given the data $\bar{X}_n^2 = 10.03702^2 = 100.742$

$$S_n^2 = 199.09634 - 100.742 = 98.3546$$

(g) Given what we learned in (d), define $a_n = \frac{n}{n+1}$, $b_n = \frac{n}{n-1}$.

Then $T_n = a_n \bar{X}_n^2$ and $W_n = b_n S_n^2$ and

$$E(T_n) = a_n E(\bar{X}_n^2) = \frac{n}{n+1} \left(\frac{n+1}{n}\right) \theta^2 = \theta^2$$

$$E(W_n) = b_n E(S_n^2) = \frac{n}{n-1} \left(\frac{n-1}{n}\right) \theta^2 = \theta^2$$

$\therefore$ both T_n and W_n are unbiased.

When $n = 10,000$, $a_n = \frac{10,000}{10,001} = 0.9999$

$b_n = 1.0001$, then $T_n = 100.732$

$W_n = 98.3644$

Also, both T_n and W_n are consistent, which is a simple application of Thm. 4.3

(h) (This exercise can be done using $\bar{X}_n^2, S_n^2$ or T_n, W_n as the variance estimator)

We will use S_n^2 . Applying Thm 4.3,

$$S_n = \sqrt{S_n^2} \xrightarrow{P} 0$$

Define $Z_n = \frac{\bar{X}_n - \theta}{\frac{S_n}{\sqrt{n}}}$, $U_n = \frac{S_n}{\theta}$,

$$V_n = \frac{Z_n}{U_n} = \frac{\bar{X}_n - \theta}{S_n / \sqrt{n}}$$

As we have seen in (c) $Z_n \xrightarrow{d} N(0,1)$

On the other hand $W_n \xrightarrow{P} 1$

so by Slutsky's Thm (4.6 iii)

we have that

$$V_n \xrightarrow{d} \frac{N(0,1)}{1} = N(0,1)$$

Thus V_n is an asymptotic pivot.

(i) The critical value for a $N(0,1)$ dist at the 1% level is $Z_{\frac{0.01}{2}}^{\text{crit}} = 2.575$.

then using V_n we get

$$\Pr\left(\bar{X}_n - 2.575 \frac{S_n}{\sqrt{n}} \leq \theta \leq \bar{X}_n + 2.575 \frac{S_n}{\sqrt{n}}\right) \approx 0.99$$

Thus, the confidence interval is

$$\left[10.03702 \pm 2.575 \frac{\sqrt{98.3546}}{\sqrt{10,000}}\right]$$

$$= [10.03702 \pm 0.255373]$$

$$= [9.78165, 10.2924]$$

(j) The true probability that we need to compute is:

$$\begin{aligned} P_r(X > 20; \theta) &= 1 - P_r(X \leq 20; \theta) = 1 - \int_0^{20} \frac{1}{\theta} e^{-\frac{x}{\theta}} dx \\ &= 1 - \left(-e^{-\frac{x}{\theta}} \Big|_0^{20} \right) = 1 + e^{-\frac{x}{\theta}} \Big|_0^{20} = 1 + \left(e^{-\frac{20}{\theta}} - 1 \right) \\ &= e^{-\frac{20}{\theta}} \end{aligned}$$

If we use $\bar{X}_n = 10.03702$ to estimate θ , we get:

$$P_r(X > 20; 10.03702) = e^{-\frac{20}{10.03702}} = 0.136337$$

Question 2

(a) β_1 is a semielasticity:

$$\Delta \% \text{ price of soda} = (100 \beta_1) \Delta \text{ proportion of blacks}$$

Since probblch is in decimals, we would read: a 1% increase in the proportion of blacks increases the price of soda by β_1 %

(b) β_2 is an elasticity:

$\Delta \% \text{ price of soda} = \beta_2 \Delta \% \text{ median fam. income}$
We would read: a 1% increase in median family income increases the price of soda by β_2 %.

(c) Using the information from the table, for the test

$$H_0: \beta_1 = 0 \text{ vs } H_a: \beta_1 \neq 0,$$

$$\text{The T statistic is } T = \frac{0.073}{0.035} = 2.08571.$$

The table indicates we would reject the null at the 5% but not 1% level.

(d) The respective details for each test:

$$\bullet H_0: \beta_2 = 0$$

vs

$$H_a: \beta_2 \neq 0$$

$$T = \frac{0.137}{0.026}$$

Table indicates we reject H_0 at 5% & 1% levels

$$\bullet H_0: \beta_3 = 0$$

vs

$$H_a: \beta_3 \neq 0$$

$$T = \frac{.390}{0.135}$$

Table indicates we reject H_0 at 5% & 1% levels

(e) The restriction matrix is $H = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}$ is the parameter vector

$$H_0: H\beta = \underline{0}_{4 \times 1} \quad \text{vs} \quad H_a: H\beta \neq \underline{0}$$

$\nearrow$ i.e. at least one of $\beta_1, \beta_2, \beta_3$ is different from zero

The test statistic: $W = (H\hat{\beta})^T (H\hat{\Omega}H^T)^{-1} (H\hat{\beta}) \sim \chi^2(3)$

W simplifies to

$$W = (\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3) S^{-1} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{pmatrix}$$

where S is the submatrix of $\hat{\Sigma}$ corresponding to variances and covariances of $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$

The observed value is $W = 39.35043$

Critical values for 5% and 1% tests are:

$$\chi_{0.05}^{\text{crit}}(3) = 7.815, \quad \chi_{0.01}^{\text{crit}}(3) = 11.345$$

Since W is larger than both we reject H_0 at the 5% and 1% levels.

(f) β_4 is an elasticity:

$\Delta\%$ price of soda = $\beta_4 \Delta\%$ median housing value.
We would read, a 1% increase in median housing value increases the price of soda by $\beta_4\%$.

The test of significance is carried out as follows:
 $H_0: \beta_4 = 0$ vs $H_a: \beta_4 \neq 0$

$T = \frac{0.121}{0.017}$. According to the table, we would reject the null at the 5% and 1% levels.

(g) The restriction matrix is $H = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

The parameter vector $\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix}$

$$H_0: H\beta = \underline{0}_{5 \times 1} \quad \text{vs} \quad H_a: H\beta \neq \underline{0}$$

$\nearrow$ i.e. at least one of $\beta_i \neq 0$ for $i = 1, 2, 3, 4$

The test statistic: $W = (H\hat{\beta})^T (H\hat{\Omega}H^T)^{-1} (H\hat{\beta}) \sim \chi^2(4)$

W simplifies to

$$W = (\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3, \hat{\beta}_4) A^{-1} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \\ \hat{\beta}_4 \end{pmatrix}$$

where A is the submatrix of $\hat{\Omega}$ corresponding to variances and covariances of $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3, \hat{\beta}_4$

The observed value of $W = 104.145$, critical values at the 5% and 1% levels are

$$\chi_{0.05}^{\text{crit}}(4) = 9.488, \quad \chi_{0.01}^{\text{crit}}(4) = 13.277. \quad \text{Since}$$

W is larger than both we reject the null at the 5% and 1% levels.

(h) Income and prpper become statistically insignificant when lhouseal is added to the regression.

(i) The restriction matrix is

$$H = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad \beta = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4)^T \text{ is the parameter vector}$$

$$H_0: H\beta = \underline{0}_{2 \times 1} \quad \text{vs} \quad H_a: H\beta \neq \underline{0}$$

$\nwarrow$ at least $\beta_2 \neq 0$ or $\beta_3 \neq 0$

The test statistic is

$$W = (H\hat{\beta})^T (H\hat{\Omega}H^T)^{-1} (H\hat{\beta}) \sim \chi^2(2)$$

$$W \text{ simplifies to } W = (\hat{\beta}_2, \hat{\beta}_3) B^{-1} \begin{pmatrix} \hat{\beta}_2 \\ \hat{\beta}_3 \end{pmatrix} \text{ where}$$

B is the submatrix of $\hat{\Omega}$ corresponding to variances and covariances of $\hat{\beta}_2, \hat{\beta}_3$

The observed value of $W = 6.56499$. Critical values at the 5% and 1% level are

$$\chi_{0.05}^{\text{crit}}(2) = 5.991, \quad \chi_{0.01}^{\text{crit}}(2) = 9.210$$

Thus we reject H_0 at the 5% level but not at the 1% level.

(j) Since model (2) contains most of the central variables, $\ln\text{seval}$ is individually significant, $\ln\text{income}$ and prpper are jointly significant at the 5% level, model (2) seems to be the most reliable.

Thus, a reasonable estimate is that if the proportion of blacks increases by 0.10, the price of soda increases by 0.98%, approx 1%.

Question 3

See solutions from 2021 Final; since it is basically the same question.